

Chenyu Zhu

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EDUCATION

Northwestern University — <i>M.S. in Robotics</i>	Sept. 2025 – Dec. 2026
University of California, Berkeley — <i>Exchange Program</i>	Jan. 2024 – Jun. 2024
Southeast University — <i>B.S. in Automation</i>	Sept. 2021 – Jun. 2025

SKILLS

Programming Languages: Python, C, C++, Assembly Language(x86), MATLAB
Skills: ROS 2, Linux, Git, CAD, SLAM, Motion Capture, Parallel Computing, Path Planning, Simulink
Machine Learning: Isaac Sim, PyTorch, Vision Language Action(VLA), Diffusion Policy
Languages: English (Fluent), Mandarin (Native), Japanese (Intermediate)

PROJECT EXPERIENCE

Memory-Augmented VLA for Long-Horizon Manipulation	Mar. 2026 – Present
Annotated dataset using OpenAI api, trained GR00T to jointly output action tokens, feeding memory back to enable persistent world state tracking, to address VLA’s failure on long-horizon tasks	
Autonomous Clothes-Folding Robot	Feb. 2026 – Present
- Collaborated to build an automated dataset generation pipeline in IsaacSim randomizing lighting, garment type, and object placement, enabling diverse manipulation training data at scale - Trained and benchmarked Action Chunking with Transformer(ACT), VLA, and Diffusion Policy against each other, achieving a best simulation success rate of 69%	
Human-in-the-Loop Compliant LiDAR UAV	Jan. 2026 – Mar. 2026
- Developed a PX4-based quadrotor using C++ that estimates human-applied external forces via a Kalman filter and responds in real time, enabling intuitive physical interaction - Implemented a potential-field obstacle avoidance controller using 2D LiDAR, allowing the UAV(Unmanned Aerial Vehicle) to navigate safely while preserving intended motion direction	
LiDAR-Based EKF SLAM from Scratch	Jan. 2026 – Mar. 2026
- Implemented EKF SLAM from scratch fusing odometry with LiDAR range-bearing data to jointly estimate robot pose and landmark positions in ROS 2 using C++, validated on a physical TurtleBot3 - Built a landmark detection pipeline using point clustering, circle-fitting, and data association, providing reliable feature extraction for the Simultaneous Localization and Mapping(SLAM) backend	
Emoji Synthesis via Pixel-Space Diffusion Model	Jan. 2026 – Mar. 2026
Implemented a PyTorch-based Diffusion Policy with a U-Net backbone, generating novel emoji compositions and demonstrating the model’s ability to capture complex visual structures	
Vision-Guided Domino Placement with Franka Emika Robot	Nov. 2025 – Dec. 2025
Built a ROS 2 pipeline using MoveIt for vision-based detection and motion planning on a Franka robot	

WORK EXPERIENCE

Robotics Research Intern – Intuitive Surgical	Shanghai, China, Jun. 2024 – Aug. 2024
Operated the da Vinci robot and analyzed thermal test data from 600+ trials, fitting cooling curves in MATLAB to quantify instrument heat dissipation	
Platform Engineer Intern – GE Healthcare	Wuxi, China, Jun. 2023 – Aug. 2023
Designed and programmed a competition robot in Python under two weeks, achieving 3rd of 10 teams	
Embedded System Intern – Cryofocus Medtech	Shanghai, China, Jun. 2022 – Sept. 2022
Programmed PD control in Embedded C for a cryogenic medical device, reducing oscillation for flow regulation precision by 12%, contributed to a patented balloon folding mechanism (CN220988900U)	